

TB Spectral Transient Shaper

DETAILED ENGLISH USER MANUAL

An STFT transient processor that separates attack and sustain by frequency bin, with Simple/EQ detection, monitor modes, saturation, clipping, limiting, and oversampling.



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Host-visible endpoints documented: 41

1. Product purpose

An STFT transient processor that separates attack and sustain by frequency bin, with Simple/EQ detection, monitor modes, saturation, clipping, limiting, and oversampling.

Broadband transient shapers treat the entire signal as one envelope, so adding punch to a kick may also exaggerate cymbals or room noise. TB Spectral Transient Shaper detects attack and sustain across individual spectral bins and allows the sensitivity curve to remain global or become frequency dependent.

What result it creates

- Frequency-dependent attack enhancement or suppression
- Independent sustain shaping
- Delta monitoring of detected and changed material
- Optional saturation, clipping, oversampling, and peak limiting

Signal flow

Input Gain -> STFT analysis -> Attack/Sustain detectors -> Simple or three-region EQ sensitivity curves -> spectral gain -> inverse STFT -> Saturation -> Clipper -> Output Gain -> Limiter

2. Complete feature guide

Attack and Sustain

Attack Gain changes fast-rising energy; Sustain Gain changes stable or decaying energy. Their independent Sensitivity and Release controls determine eligibility and recovery.

Simple Mode

One horizontal sensitivity value applies across the whole spectrum for Attack and another for Sustain. It is the fastest way to establish whether the process fits the source.

EQ Mode

Independent Low, Mid, and High frequency, Q, and sensitivity nodes build frequency-dependent detector curves for Attack and Sustain.

Monitor modes

Normal, Normal Delta, Attack, Attack Delta, Sustain, and Sustain Delta reveal either the full result or the material identified/changed by a particular detector.

Saturation

Enable Saturation, choose Soft, Punch, Warm, Digital, Fold, or Hard, set Drive, and optionally link the shaper behavior through Saturation Shaper Link.

Clipper and Limiter

Clipper uses its own threshold for hard peak shaping. Limiter uses Limiter Threshold and the Lookahead control. They are separate final safety/character stages.

Oversampling

x1, x2, x4, or x8 affects the non-linear finishing stages and CPU/latency tradeoff. Choose higher factors when distortion quality matters.

3. Quick start

1. Start in Simple Mode with Saturation, Clipper, and Limiter off.
2. Lower Attack Sensitivity until desired hits react, then set Attack Gain.
3. Lower Sustain Sensitivity until body or tails react, then set Sustain Gain.
4. Tune both Release controls.
5. Switch to EQ Mode only when the reaction must be frequency selective.
6. Use Monitor modes to confirm the target.
7. Add finishing stages last and level-match Input/Output.

4. Practical workflows

Tighter drum room

1. Reduce Sustain Gain.
2. Set Sustain Sensitivity to catch the room tail.
3. Use EQ Mode to leave low sustain intact if needed.
4. Monitor Sustain Delta to confirm what is removed.

Expected result: Room decay shortens while the direct hit remains focused.

More kick punch

1. Raise Attack Gain.
2. In EQ Mode focus Attack Low/Mid sensitivity around the kick impact.
3. Use a short-to-medium Attack Release.
4. Enable the limiter only if peaks need control.

Expected result: Impact increases in the intended band without equally lifting cymbals.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- STFT processing introduces latency and can smear extremes at aggressive settings.
- Monitor modes are diagnostic and must be returned to Normal.
- Clipper and Limiter can conceal excessive transient gain; set the shaper first.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 41 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
ATK_HIGH Frequency VST3 ID 1515743726	20.00 to 20000.00	8000.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
ATK_HIGH Q VST3 ID 1015367835	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
ATK_HIGH Sensitivity VST3 ID 1516118797	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
ATK_LOW Frequency VST3 ID 700840586	20.00 to 20000.00	100.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
ATK_LOW Q VST3 ID 1629937023	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
ATK_LOW Sensitivity VST3 ID 701215657	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
ATK_MID Frequency VST3 ID 869642262	20.00 to 20000.00	1000.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
ATK_MID Q VST3 ID 1630663539	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
ATK_MID Sensitivity VST3 ID 870017333	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
Attack Gain VST3 ID 1091520742	-24.00 to 24.00	4.00	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Attack Release VST3 ID 1586716960	1.0 to 2000.0	50.0	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Attack Sensitivity VST3 ID 1091882238	0.00 to 1.00	0.50	Automatable	Sets how quickly the associated detector, envelope, grain, or dynamics stage responds at onset.
Bypass VST3 ID 121080692	Off, On	Off	Automatable	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Clipper VST3 ID 220334386	Off, On	Off	Automatable	Controls or enables non-linear density, harmonic color, or peak shaping.
Clipper Threshold VST3 ID 374122012	-20.00 to 0.00	-0.70	Automatable	Sets the level at which the associated detector, dynamics stage, clipper, or limiter begins to act.
Input Gain VST3 ID 538763865	-24.00 to 24.00	0.00	Automatable	Sets the level entering the processor and therefore changes downstream drive or detection.
Limiter VST3 ID 1289122898	Off, On	Off	Automatable	Host-automatable control for Limiter. Its full range and default are shown in this table; use it with the related feature section above.
Limiter Threshold VST3 ID 982698300	-24.00 to 0.00	-0.10	Automatable	Sets the level at which the associated detector, dynamics stage, clipper, or limiter begins to act.
Lookahead VST3 ID 1770736226	0.00 to 20.00	0.00	Automatable	Provides future-signal context to the associated dynamics or limiter stage and may add latency.
Mode VST3 ID 417437801	Simple, EQ	Simple	Automatable	Selects the operating algorithm, response shape, or tonal character for the named block.
Monitor VST3 ID 1931564424	Normal, Normal Delta, Attack, Attack Delta, Sustain, Sustain Delta	Normal	Automatable	Host-automatable control for Monitor. Its full range and default are shown in this table; use it with the related feature section above.
Output Gain VST3 ID 866227696	-24.00 to 24.00	0.00	Automatable	Sets or limits the final output level after the plug-in's main processing.
Oversampling VST3 ID 196524042	x1, x2, x4, x8	x1	Automatable	Selects the internal oversampling factor, trading more CPU/latency for cleaner non-linear processing.

Parameter	Range / options	Default	Host status	Function
Program VST3 ID 1886548852	Init, Snap Kick, Room Kill, Big Room, Top End Crack, Sub Punch, Parallel Slam, Finger Definition, Sub Hold, Amp Grit, Pick Attack, Soft Strum, Piano Bloom, Metallic Edge, Consonant Control, Breath Forward, Vocal Presence, Glue Tighten, Mix Punch, Open The Mix, Master Safety, Reverse Swell, Gated Spikes, Frozen Tail, Broken Transients	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Saturation VST3 ID 1252710312	Off, On	Off	Automatable	Sets the strength of the associated dynamics response.
Saturation Drive VST3 ID 1944639409	0.00 to 36.00	0.00	Automatable	Sets the strength of the associated dynamics response.
Saturation Mode VST3 ID 825005692	Soft, Punch, Warm, Digital, Fold, Hard	Soft	Automatable	Sets the strength of the associated dynamics response.
Saturation Shaper Link VST3 ID 692104358	Off, On	Off	Automatable	Sets the strength of the associated dynamics response.
SUS_HIGH Frequency VST3 ID 1315314279	20.00 to 20000.00	8000.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
SUS_HIGH Q VST3 ID 1023218370	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
SUS_HIGH Sensitivity VST3 ID 1315689350	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
SUS_LOW Frequency VST3 ID 1802753777	20.00 to 20000.00	100.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
SUS_LOW Q VST3 ID 1560916600	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
SUS_LOW Sensitivity VST3 ID 1803128848	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
SUS_MID Frequency VST3 ID 1971555453	20.00 to 20000.00	1000.00	Automatable	Sets the principal frequency, note, or spectral center used by this function.
SUS_MID Q VST3 ID 1561643116	0.10 to 10.00	1.00	Automatable	Sets bandwidth or resonance sharpness for the named filter or analysis band.
SUS_MID Sensitivity VST3 ID 1971930524	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.
Sustain Gain VST3 ID 1366388941	-24.00 to 24.00	-2.00	Automatable	Sets gain for the named input, output, band, or parallel send/return path.
Sustain Release VST3 ID 1830083545	1.0 to 2000.0	50.0	Automatable	Sets how long the associated detector, envelope, reverb, or resonant process takes to recover or decay.
Sustain Sensitivity VST3 ID 1366750437	0.00 to 1.00	0.50	Automatable	Sets how sensitively the associated analysis or transfer stage responds to source detail.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.